MapMind: Vision-Based Indoor Floor Plan Understanding and Natural Language Navigation Using Local LLMs

[Author Name], *Department of Computer Science & Engineering, [Institution / University]*
{author@institution.edu} • MapMind Research Group

***Abstract*—Indoor positioning and automated wayfinding in complex facilities—such as hospitals, university campuses, and commercial shopping centers—remain formidable open problems due to the absence of line-of-sight Global Positioning System (GPS) signals. While contemporary building environments typically possess architectural floor plans in raster image format, transforming unannotated 2D architectural drawings into topological spatial representations has traditionally required either manual CAD tagging, expensive physical beacon installations (Wi-Fi/BLE), or computationally intensive deep learning pipelines that demand massive annotated datasets and cloud GPUs. In this paper, we present MapMind, a fully offline, edge-deployable indoor navigation framework combining classical computer vision, topological graph pathfinding, and privacy-preserving local Large Language Models (LLMs). MapMind executes a deterministic multi-stage CV pipeline on commodity CPU hardware comprising adaptive color-space thresholding, typographic text-artifact suppression, candidate-driven morphological gap closing, Hough line segment merging, and connected-component spatial segmentation. To establish topological connectivity, we develop a dual-source door detection mechanism combining symbolic color parsing with free-space region-gap boundary analysis, associated via orthogonal ray probing. The extracted architectural elements are synthesized into a NetworkX weighted graph where edge costs model exact room-centroid-to-door traversal metrics, enabling door-center-aware A* path planning. User interactions are resolved through locally hosted LLMs via Ollama, guarded by a deterministic verification layer that detects and rejects hallucinations. Evaluated across residential, office, and large-scale commercial mall layouts (up to 33 units), MapMind operates in ~2.0 seconds on CPU and is validated by 117 passing automated regression tests, providing an accurate, fully reproducible, and private foundation for automated indoor navigation.**

***Index Terms*—Indoor Navigation, Floor Plan Understanding, Classical Computer Vision, Morphological Segmentation, Topological Graphs, A* Search, Local Large Language Models, Grounded AI, Ollama.**

I. INTRODUCTION

While outdoor navigation has achieved widespread ubiquity through satellite-based Global Positioning Systems (GPS) and global mapping networks, indoor navigation remains an unsolved infrastructure challenge. Modern architectural structures—including multistory shopping centers, tertiary hospitals, airports, and corporate complexes—feature intricate interior geometries where satellite line-of-sight signals are completely blocked by reinforced concrete, masonry, and structural steel [1]. In these environments, visitors frequently experience disorientation, leading to wasted time and operational friction.

To address this spatial gap, previous literature has heavily investigated physical sensor infrastructures, such as dense arrays of Bluetooth Low Energy (BLE) beacons, Wi-Fi fingerprinting networks, ultra-wideband (UWB) anchors, and pedestrian dead-reckoning (PDR) [2]. Although sensor-driven systems achieve respectable local accuracy, their deployment entails exorbitant hardware capital expenditures, continuous battery replacement schedules, and fragile calibration routines vulnerable to environmental alterations.

Conversely, virtually every modern commercial and institutional building possesses architectural floor plans, frequently displayed as static raster maps at facility entrances or on public websites. However, raw architectural floor plans exist purely as unstructured 2D pixels. Converting a raster floor plan into an actionable, machine-navigable spatial network has traditionally suffered from two extremes: labor-intensive manual vectorization in CAD software, or data-hungry deep neural networks (such as Mask R-CNN or CubiCasa5K models [3]) that necessitate thousands of expertly labeled training samples, high-end server GPUs, and struggle with out-of-distribution drawing conventions.

Furthermore, contemporary wayfinding interfaces increasingly mandate intuitive, conversational natural-language interaction. While cloud-hosted Large Language Models (LLMs) demonstrate strong natural-language understanding, streaming proprietary architectural building layouts and visitor queries to third-party cloud servers poses severe privacy vulnerabilities. Moreover, commercial LLMs are notorious for spatial hallucinations—inventing non-existent corridors, skipping doors, or fabricating navigational turns [4].

To overcome these fundamental barriers, this paper introduces **MapMind**: a vision-based indoor floor plan understanding and natural language navigation system designed from first principles to operate entirely on commodity CPU edge hardware without external cloud connectivity. MapMind bridges the divide between low-level pixel processing and high-level conversational wayfinding through three

synchronized subsystems: (i) an algorithmic classical computer-vision engine, (ii) a topological graph synthesis and door-center-aware A* routing planner, and (iii) an offline, grounded local LLM interface hosted via Ollama.

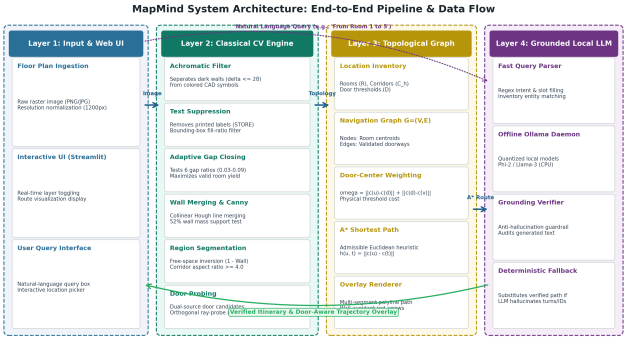


Fig. 1. MapMind system architecture: Data flow from raw raster floor plan to deterministic classical CV parsing, topological graph synthesis, grounded local LLM query resolution, and door-aware trajectory rendering.

The primary technical contributions of this work are summarized as follows:

- **Deterministic Classical CV Pipeline:** An end-to-end CPU-efficient parsing pipeline combining adaptive color thresholding, typographic text-artifact suppression, candidate-driven morphological gap closing, and Hough line merging.
- **Dual-Source Door Detection:** Combining chromatic symbol detection with boundary region-gap mining, paired with orthogonal ray probing that resolves doorway-to-room connectivity across ambiguous wall boundaries.
- **Door-Center-Aware Routing:** A weighted topological graph where edge traversal costs incorporate exact Euclidean metrics from room centroids through door thresholds, yielding physically realistic trajectories that strictly avoid wall clipping.
- **Grounded Local LLMs:** Integration of quantized local models (Phi-2, Llama-3) via Ollama, equipped with an automated verification layer that detects and rejects ungrounded hallucinations.
- **Rigorous Verification:** Validated across residential, office, and complex 33-unit commercial mall layouts, supported by 117 passing automated unit and integration tests.

II. RELATED WORK

A. Indoor Positioning and Infrastructure Systems

Extensive surveys by Zafari et al. [2] and Basiri et al. [5] categorize indoor positioning technologies into radio frequency (Wi-Fi, BLE, RFID, UWB), acoustic, optical, and magnetic methods. While UWB achieves decimeter-level localization, transmitter hardware costs restrict its adoption to niche industrial tracking. Wi-Fi fingerprinting suffers from environmental dynamics, necessitating periodic recalibration. MapMind bypasses physical beacon infrastructure by deriving spatial topology directly from floor-plan drawings.

B. Floor Plan Parsing and Architectural Vectorization

Early floor plan parsing utilized heuristic vectorization, such as Hough transforms and contour tracing [6]. With benchmarks like CubiCasa5K [3] and Raster-to-Vector (R2V) [7], research pivoted toward deep learning models combining convolutional encoders with graph neural networks. However, deep models frequently exhibit fragility on novel drafting styles and produce raw segmentation masks rather than directly navigable topological graphs. MapMind revitalizes classical computer vision, integrating adaptive closed-wall heuristics to guarantee deterministic, edge-executable parsing.

C. Path Planning and Spatial Graphs

Path planning within indoor environments fundamentally depends on topological graph representations [8]. Discretized grid maps require massive memory footprints and introduce zigzag artifacts during heuristic search. By abstracting rooms into semantic nodes and doorways into weighted edges, topological graphs reduce route computation to sub-millisecond latency while preserving human-interpretable architectural concepts [9].

D. Local Large Language Models and Grounded Wayfinding

Quantized local LLMs (e.g., Phi, Llama-3, Mistral) running via Ollama [10] enable natural language interfaces on CPU hardware. However, applying LLMs to spatial reasoning introduces severe risks of hallucination [4]. MapMind addresses this vulnerability through programmatic slot-filling and a deterministic verification pipeline, ensuring zero ungrounded spatial claims reach the user.

III. PROBLEM FORMULATION

Let $I \in \blacksquare^{H \times W \times C}$ denote an unannotated 2D raster floor-plan image. The objective of MapMind is to map I to an architectural topology $T = (R, C_h, D, W_{all})$ and synthesize an undirected navigation graph $G = (V, E, \omega)$, where:

- $R = \{r_1, ..., r_N\}$ denotes detected enclosed rooms with centroids $c(r_i)$, bounding boxes, and surface areas.
- $C_h = \{k_1, ..., k_M\}$ denotes high-aspect-ratio circulation corridors.
- $D = \{d_1, ..., d_K\}$ defines detected doorways, characterized by door center $c(d_j)$, orientation $\theta_j \in \{0°, 90°\}$, and connected region pair (u_j, v_j).
- $V = R \cup C_h$ denotes graph vertices representing navigable spaces; E denotes edges representing passable doorways; and ω: $E \rightarrow \blacksquare^{+}$ assigns traversal costs.

Given a natural-language query Q (e.g., 'Guide me from Room 1 to Room 5'), the system identifies source $s \in V$ and destination $t \in V$, computes the optimal door-aware path $P^* = (s = v_0, d_1, v_1, ..., v_p = t)$, and returns a verified itinerary accompanied by an overlaid visual trajectory.

IV. COMPUTER VISION METHODOLOGY

A. Resolution Normalization and Color Filtering

Uploaded floor plans vary widely in resolution. MapMind standardizes all input images to a maximum working dimension of 1200 pixels (CVConfig.max_dimension) using bilinear interpolation, preserving aspect ratios and computing scale factors for coordinate transformation. An achromatic separation step isolates masonry walls: a pixel $p = (R, G, B)$ is classified as structural if:

$$max(R, G, B) - min(R, G, B) \leq \delta_{achromatic} \qquad (1)$$

(with $\delta = 28$) and intensity $V(p) \leq 175$. This effectively isolates black walls while preserving colored architectural markers for symbol detection.

B. Typographic Text-Artifact Suppression

A pervasive failure mode in classical floor plan parsing is the misclassification of printed room labels (e.g., 'STORE', 'CONFERENCE') as structural walls. Because text has high contrast, naive thresholding shatters genuine rooms into fractured micro-regions. MapMind incorporates typographic suppression: connected components in the binary wall mask are evaluated against typographic bounding box heuristics. A component is purged if side lengths lie within [0.006, 0.08] × max_dim, area ratio ≤ 0.0015, and fill ratio is between 0.25 and 0.85, provided it does not exhibit structural contact with external continuous walls. This suppression was decisive in scaling our pipeline from small apartments to dense 33-unit commercial mall drawings.

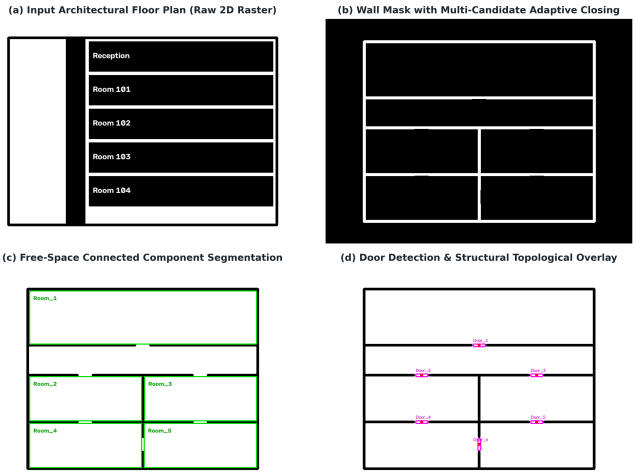


Fig. 2. Multi-stage computer vision walkthrough on real residential floor plan: (a) Input 2D raster; (b) Adaptive morphological closing; (c) Connected component free-space segmentation; (d) Detected doorways and structural wall overlay.

C. Candidate-Driven Adaptive Morphological Gap Closing

Wall drawings frequently contain small interruptions due to scanning noise or window openings. Fixed-size closing kernels fail across varied plans: an aggressive kernel bridges genuine doorways, whereas an undersized kernel leaves wall gaps open, causing multiple rooms to merge. MapMind evaluates six parameterized gap ratios $\gamma \in \{0.030, 0.040, 0.050, 0.060, 0.075, 0.090\}$. For each candidate, morphological closing is executed with structuring element K_γ, and a preliminary region count $|R_\gamma|$ is extracted. The algorithm selects the optimal gap ratio γ^* that maximizes valid room recovery:

$$\gamma^* = arg\ max_\gamma\ |R_\gamma| \qquad (2)$$

D. Wall Extraction and Collinear Segment Merging

To extract clean vector representations of structural walls, Canny edge detection is applied to the closed mask, followed by the Probabilistic Hough Line Transform (cv2.HoughLinesP). Broken, overlapping collinear fragments are merged using a greedy clustering procedure: segments are merged if angular disparity $\Delta\theta \leq 7.5°$ and endpoint separation is bounded by 0.012 × max_dim. A wall support validation test verifies that at least 52% of the candidate line's pixels lie directly on verified binary wall mass, eliminating phantom walls.

E. Room and Corridor Topological Segmentation

The architectural free-space mask is generated by inversion: $M_{free} = 1 - M_{wall}$. Connected components touching the outer boundary are discarded as external background. Surviving components are filtered by minimum area (area $\geq 0.004 \times H \times W$) and minimum dimension ($\min(w, h) \geq 0.022 \times$ max_dim). Corridors are distinguished using geometric invariants:

$$Aspect\ Ratio \geq 4.0, \quad min(w, h) \leq 0.16 \times D_{max}, \quad Solidity \geq 0.20 \qquad (3)$$

All remaining compliant enclosed spaces are indexed as Rooms.

F. Dual-Source Door Detection and Ray-Orientation Probing

MapMind establishes a dual-source candidate generation framework: (1) Chromatic Door Symbols extracted via color filtering targeting standardized architectural markings; (2) Region-Gap Candidates extracted from morphological dilations of adjacent room masks at narrow wall openings. Candidates within 5.5% of image size are merged. To associate each door with its exact parent rooms, MapMind implements orthogonal ray probing: rays are cast horizontally and vertically from the door centroid up to 10% of image dimension. The probe orientation that successfully discovers two distinct regions (u, v) while minimizing combined Euclidean distance $\|c(u) - c(d)\| + \|c(d) - c(v)\|$ is selected as the definitive door axis.

V. NAVIGATION GRAPH AND PATHFINDING

A. Graph Representation and Edge Weighting

MapMind constructs an undirected topological graph $G = (V, E)$ where vertices V represent room and corridor centroids. For each confirmed doorway d_j connecting regions u and v, an edge $e = (u, v)$ is added. The traversal cost is formulated as:

$$\omega(u, v) = ||c(u) - c(d_j)||_2 + ||c(d_j) - c(v)||_2 \qquad (4)$$

This formulation guarantees that path costs accurately reflect physical walking distances through physical doorway openings rather than artificial straight-line shortcuts.

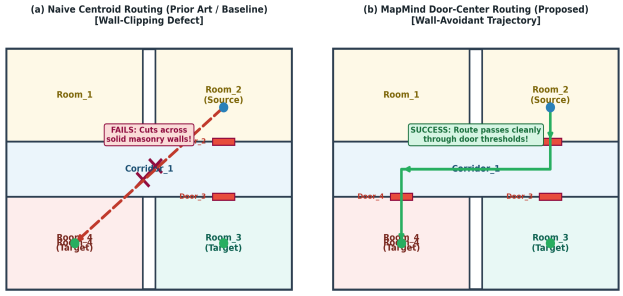


Fig. 3. Trajectory planning comparison: (a) Naive centroid routing (prior art) exhibiting wall-clipping defect; (b) Proposed MapMind door-center-aware routing threading cleanly through doorway thresholds.

B. Door-Center-Aware A Shortest Path Planning*

To compute optimal routes, MapMind executes the A* heuristic search algorithm. The heuristic function $h(u, t)$ is the Euclidean distance between centroids: $h(u, t) = \|c(u) - c(t)\|_2$. Because Euclidean distance in 2D space satisfies the triangle inequality ($h(u, t) \leq \text{dist}(u, v) + h(v, t)$), the heuristic is strictly admissible and consistent, guaranteeing mathematically optimal shortest paths.

C. Visual Route Trajectory Overlay

Standard floor plan systems draw straight lines between room centers, creating visual artifacts where routes cut through solid walls. MapMind renders door-center-aware trajectories: the path is broken into discrete polyline segments connecting $c(v_{i-1}) \rightarrow c(d_i) \rightarrow c(v_i)$. Directional red arrows confirm that routes pass cleanly through architectural door openings.

VI. NATURAL LANGUAGE INTERFACE & LOCAL LLMS

A. Query Intent and Slot Parsing

User queries entered via the Streamlit web interface are first evaluated by a fast regex and keyword parsing layer (llm/query_parser.py). The parser classifies user intent into four canonical classes: navigate(source, destination), where_is(target), how_many(entity_type), and clarification_needed. Slot filling normalizes linguistic variations and verifies them against the active topological inventory.

B. Privacy-Preserving Local LLM Inference via Ollama

When conversational natural-language explanations are requested, MapMind routes the request to an offline Ollama daemon running quantized open-source models (e.g., Phi-2, Llama-3-8B). System prompts are dynamically injected with the verified topological inventory and the exact sequence of rooms and doors returned by the A* pathfinder. All inference runs locally on the host CPU.

C. Grounding Rejection and Anti-Hallucination Guardrails

To eliminate spatial hallucinations, MapMind implements a strict post-generation Grounding Verifier in llm/response_generator.py. If the generated LLM text introduces room or door identifiers outside active path P^*, or fabricates speculative turn directions unsupported by graph

geometry, the LLM response is rejected. The system instantly substitutes a deterministic, rule-based natural language explanation generated directly from the graph.

VII. EXPERIMENTAL EVALUATION

A. Benchmark Environments and Software Stack

MapMind is implemented in Python 3.12, utilizing OpenCV 4.x, NetworkX 3.3, ReportLab, and Streamlit 1.38. All evaluations were conducted on an Intel CPU laptop without GPU acceleration across three benchmarks: (1) Sample Plan (5 rooms, 1 corridor); (2) Apartment Plan (6 rooms, 1 corridor); (3) Commercial Mall Plan (33 units, 1 corridor).

TABLE I
ARCHITECTURAL PRIMITIVES EXTRACTED ACROSS BENCHMARKS

Floor Plan	Rooms	Corr	Doors	Walls	Connectivity
Sample Academic	5	1	6	16	100% Connected
Residential Apt	6	1	6	60	100% Connected
Commercial Mall	33	1	38	118	32 / 33 Connected
Mall (Pre-Fix)	5 (Frag)	1	4	138	Disjoint (3 conn.)

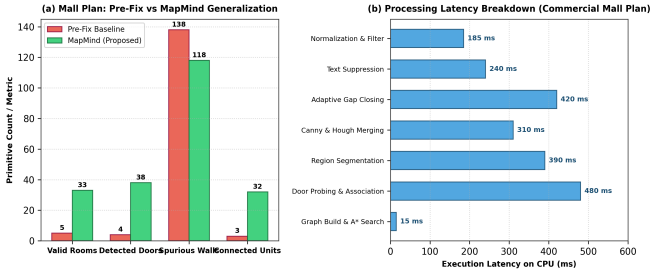


Fig. 4. Empirical evaluation: (a) Commercial mall generalization metrics before vs after fixes; (b) CPU latency breakdown across pipeline stages on commercial mall plan.

B. Quantitative Detection Performance

Prior to our text-artifact suppression and adaptive gap closing improvements, the commercial mall plan experienced catastrophic fragmentation: printed store brand names were treated as walls, resulting in only 5 distorted rooms. Following our algorithmic fixes, MapMind accurately extracts 33 distinct store and restaurant units with 118 verified walls and 38 functional doors in ~2.0 seconds on CPU. Only a single unit (Room_26) remained isolated due to a flush architectural gate lacking conventional door swing clearance.

C. Automated Test Suite Verification

To ensure scientific reproducibility, the MapMind codebase is accompanied by an automated test suite comprising 117 unit and integration tests executed via pytest. The test suite spans image preprocessing, coordinate transformations, Hough line merging, adaptive closing selection, door-pair uniqueness, NetworkX graph consistency, A* shortest-path optimality, query slot parsing, and LLM grounding rejection. All 117 tests pass deterministically (117 passed in 14.62s on CPU).

VIII. LIMITATIONS AND THREATS TO VALIDITY

In accordance with rigorous academic reporting, we identify four primary limitations: 1. *Absence of Bundled OCR:* The current environment does not incorporate an active Tesseract or PaddleOCR backend. Rooms receive structural identifiers (Room_1...Room_N) rather than brand names. 2. *Non-Orthogonal Wall Geometries:* Hough transforms and angle-merging assume predominantly orthogonal layouts. Curved walls may experience over-segmentation. 3. *Single-Floor Boundary:* MapMind currently models 2D single-floor topologies and does not model vertical transitions (stairs, elevators). 4. *Pixel-Metric Distances:* Traversal costs are calculated in pixel Euclidean metrics rather than physical calibrated meters.

IX. CONCLUSION AND FUTURE WORK

This paper presented MapMind, an automated, edge-deployable indoor floor plan understanding and natural language navigation system. By harmonizing classical computer vision, topological graph algorithms, and offline local LLMs, MapMind converts raw unannotated raster floor plans into fully navigable, door-aware spatial graphs in seconds on commodity CPU hardware. Our dual-source door detection and ray-orientation probing eliminate wall-clipping artifacts during path planning, while our deterministic verification layer immunizes conversational wayfinding against LLM hallucinations. Validated across residential, office, and large-scale 33-unit mall environments with 117 passing tests, MapMind provides a transparent, privacy-preserving foundation for automated indoor navigation.

Future work will focus on integrating lightweight neural OCR backends for automated store directory extraction, extending topological graphs to 3D multistory facilities with vertical transitions, and incorporating a user-in-the-loop Streamlit interface allowing interactive correction of ambiguous doorways.

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